



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

QUANTUM ALGORITHMS AND COMPLEXITY [CSE-40B2]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain how controlled-phase gates in QFT affect entanglement between qubits. What happens if you remove some of them? (5)
 - A)
 - B) Design a quantum circuit that prepares the 3-qubit ABC state $|ABC\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$. Explain the steps. (5)
- 2) Explain why is Deutsch-Jozsa Algorithm is exponentially faster than any classical deterministic algorithm for distinguishing between constant and balanced functions? (5)
 - A)
 - B) A quantum developer decides to use Grover's algorithm on a problem with a known structure (e.g., sorted data). Analyze whether Grover's algorithm is optimal in this context. What alternatives might be more efficient? (5)
- 3) Given a unitary operator U and an eigenvector $|\psi\rangle$ such that $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, describe how the Quantum Phase Estimation (QPE) algorithm estimates θ . (5)
 - A)
 - B) Evaluate the plausibility of BQP containing NP-complete problems and the implications such a discovery would have for complexity theory. (5)
- 4) Explain how Shor's algorithm uses order finding to factor a composite number N , and why this is exponentially faster than classical methods. (5)
 - A)
 - B) Explain why does Quantum Counting not require the exact number of Grover iterations, unlike standard Grover's algorithm? Analyze the role of phase estimation? (5)
- 5) Evaluate the efficiency of the Bernstein-Vazirani algorithm compared to classical approaches for finding the hidden string. (5)
 - A)
 - B) Propose a real-world problem domain where quantum advantage could have a transformative impact, and justify its feasibility based on complexity class analysis. (5)

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